

SUJAL DEB

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Professional Summary

Data Analyst with proven experience building end-to-end analytics pipelines across 15M+ record datasets – leveraging SQL, Python and machine learning to surface insights that drive business decisions. Skilled in translating complex statistical outputs into stakeholder-ready narratives that influence cross-functional strategy. Experienced in credit risk modeling, vendor performance analytics, ETL pipeline automation, and delivering executive dashboards through Power BI. Focused on financial and transactional analytics with a consistent emphasis on business impact, data integrity, and decision-enabling insight.

Technical Skills

- **Programming & Data Analysis:** Python (Pandas, NumPy, SciPy, Scikit-learn), SQL (CTEs, Window Functions, Indexing, Query Optimization).
- **Databases:** MySQL, SQLite, Data Cleaning, ETL Pipelines.
- **Machine Learning & Statistics:** Regression, Classification (Random Forest), Clustering, Hypothesis Testing, Feature Engineering.
- **Data Visualization:** Power BI, Tableau, Matplotlib, Seaborn, MS Excel
- **Deployment & Cloud:** Google Cloud, Streamlit, BigQuery, Model Serialization (Pickle, Joblib)

Experience

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Jan 2025 – Dec 2025

Data Analyst Intern, Noida, India

- Automated large-scale lead data extraction (22 attributes) using Python and Selenium, eliminating 90% of manual processing effort and freeing team bandwidth for higher-value analysis – reducing data inconsistency errors by 95% post-migration into MySQL.
- Translated raw lead pipeline data into structured analytical reports and visual dashboards, communicating funnel inefficiencies and high-performing segments to non-technical marketing stakeholders to drive actionable channel decisions.
- Identified underperforming acquisition domains through end-to-end pipeline analysis and presented findings to the marketing team – directly redirecting budget focus toward high-converting segments and driving a 40% improvement in conversion rate.
- Owned lead volume KPIs as the analytical point of contact across teams – synthesizing data signals into strategic recommendations that increased lead volume by 30%, demonstrating ability to influence cross-functional decisions through data-backed narrative.

Projects

Jan 2026

Vendor Segmentation & Performance Analysis, Independent

- Designed a **chunked ETL pipeline** ingesting a 1.4GB **dataset (15M+ records)** into SQLite, **optimizing SQL CTEs** and strategic indexing to **reduce query runtime by 48%** and eliminate manual reporting overhead – delivering a fully automated, **repeatable analytics pipeline**.
- Conducted **Mann-Whitney hypothesis** testing and **Pareto analysis** revealing **13.5% of vendors drive 80%** of revenue while **83% over-purchased in 2024** – translating statistical outputs into **procurement strategy recommendations** for leadership.
- Built a **K-Means clustering** model segmenting 10,500+ vendor-brands into **4 performance tiers**, directly informing inventory investment decisions by identifying that 100% of **underperformance risk concentrates in a single vendor segment**.
- Trained a **Random Forest classifier** (98% accuracy, 0.987 ROC-AUC) to proactively **flag vendor risk** before capital exposure occurs, delivered through a **multi-page Power BI dashboard** with executive **KPI cards** designed for non-technical procurement stakeholders

Credit Risk & Loan Amount Prediction, Independent

Dec 2024

- Built a production-realistic credit risk **pipeline on 2.26M** LendingClub records using strictly **application-time features** – enforcing **removal of 16 post-issuance** leakage columns to ensure model outputs could actually **inform real lending** decisions, not just benchmark scores.
- Validated all predictors through **hypothesis testing (T-test, Chi-Square, ANOVA)** before model training, translating statistical **findings into business narrative**: defaulters **earn \$7,226 less on average** and loan grade predicts default monotonically from **7% to 55.4%**.
- Trained a **LightGBM classifier** (0.774 ROC-AUC, 68% recall) **identifying \$417M in default exposure** across test records – prioritizing **recall as the primary metric** because a missed defaulter **represents a direct capital loss**, not just a model error.
- Engineered 5 **domain-driven financial features (loan-to-income ratio, installment burden, credit utilization risk)** and deployed **predictions via Streamlit** – enabling non-technical stakeholders to **query borrower risk at the point of application decision**.

Education

Lovely Professional University

Bachelor Of Technology in Computer Science and Engineering, **CGPA: 8/10**

2020-2024

Punjab, India

Coursework: Engineering Mathematics, DBMS, DSA, AI/ML, Probability and Statistics, Virtualization and Cloud Computing, Data Visualization, OS, Computer Network and Design, Industry Ethics.

Certification

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| • Data Science Certification with GenAI and Machine Learning, Career247. | Ongoing |
| • Data Science and Machine Learning Internship Program, Edureka. | May - 2024 |
| • Database Management Systems, Infosys Springboard. | Jul - 2023 |